MATH102BKK · HARBOUR.SPACE BANGKOK

Intro to Higher Math

From getting the answer to knowing why the answer is right.

Session	Date	Today
1 of 15	Mon 7 Sep 2026	Overview + refresher

INTRODUCTIONS

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Software Engineers at Autopilot — and we have done all of this together.

2017 – 2023
School №199, Tbilisi
Physics & Mathematics

since 2022
Teaching mathematics
Privately, then here

2023 – 2026
Harbour.Space @ UTCC
CS & Data Science, double major

since 2023
ICPC
Competing for Harbour.Space

since 2025
Brainstormers
Co-founded — tournaments in logic

2026
Autopilot
Fintech and AI systems

Who you are

1

What is your name?

And what you would like to be called in class.

2

Where are you from?

Country, city — and how long you have been in Bangkok.

3

What is your background in maths?

Last course you took, and how you feel about proofs.

WHY WE ASK This class runs on discussion and group work. Knowing the room — and hearing that other people are also new to proof — is what makes people willing to speak.

WHY THIS COURSE EXISTS

School maths ends where university maths begins

What you were rewarded for

- → Getting the number at the bottom of the page
- → Recognising which technique to apply
- → Speed and accuracy

What you will be rewarded for here

- → A claim, stated precisely
- → An argument that forces the claim to be true
- → Writing it so another person is convinced

THE SHIFT A calculation can be checked. An argument must be **understood**. Everything in the next three weeks trains the second skill.

LOGISTICS

The course at a glance

3
weeks · 7–25 Sep 2026

45
contact hours · 3 h/day

4
ECTS credits

Where it takes you next

Calculus · Linear algebra · Discrete mathematics · Graph theory · Probability — every one of them assumes the language you build here.

Prerequisites

None formally. A solid high-school background is assumed, and today's second half checks it is actually there.

Books

Simmons, *Precalculus Mathematics in a Nutshell* · Barrington, *A Mathematical Foundation for Computer Science* · Gibilisco, *Algebra Know-It-All*

ROADMAP

The course, week by week

Week 1 · Language

1. Overview + refresher
2. Sets & set operations
3. Functions
4. Function families
5. Proof techniques

Week 2 · Method

6. Mathematical induction
7. Combinatorics + binomial theorem
8. **Midterm**
9. GCD & divisibility
10. Modular arithmetic

Week 3 · Structure

11. Vectors in $\mathbb{R}^n$
12. Polynomials
13. Complex numbers
14. Recap · maths in the AI era
15. **Final exam**

How the grade is built

COMPONENT	WEIGHT
Homework assignments	60%
Final exam	20%
Midterm exam	10%
Participation & in-class activities	10%

Homework dominates deliberately: proof is a skill, and skills are built by repetition, not by one exam morning.

MARKING RULE

A correct final answer with no reasoning is worth **very little**.

A wrong final answer with a clear, nearly-correct argument is worth **a lot**.

GROUND RULES

Collaboration, AI, and the one hard line

ENCOURAGED Discuss the material. Compare strategies. Argue about solutions. Use AI tools as a learning aid.

PROHIBITED Copying solutions from other students or external sources and submitting them as your own.

THE TEST THAT SETTLES IT You must be able to **explain and justify every solution you submit**, on the spot, without notes. If you cannot, it is not yours yet — whoever or whatever produced it.

PART II

School math refresher

Not revision. We take four things you already compute and ask what makes them true — which is the habit the rest of the course is built on.

2

NOTATION

The language we will use

SYMBOL	READS AS
$\in$	is an element of
$\subseteq$	is a subset of
$\implies$	implies
$\iff$	if and only if
$\forall$	for all
$\exists$	there exists
$\mathbb{N}\,\mathbb{Z}\,\mathbb{Q}\,\mathbb{R}$	naturals, integers, rationals, reals
$\sum_{k=1}^{n}$	the sum over $k = 1$ to n

DEFINITION $P \implies Q$ says: whenever P is true, Q must be true too. The only way it can fail is if P is true and Q is false.

TRAP $P \implies Q$ is not $Q \implies P$. "$x = 2 \implies x^2 = 4$" is true; its converse fails at $x = -2$.

Rules come with conditions

LAWS OF EXPONENTS For $a > 0$ and all real m, n:

$$a^m a^n = a^{m+n} \qquad (a^m)^n = a^{mn} \qquad a^{-n} = \frac{1}{a^n}$$

WHY $A^0 = 1$ Put $n = 0$ in the first law: $a^m a^0 = a^m$. As $a \neq 0$ we may divide by a^m, giving $a^0 = 1$. It is forced by the law, not chosen for convenience. ∎

TRAP $\sqrt{x^2} = x$ is false. The radical returns the non-negative root, so $\sqrt{x^2} = |x|$. Test $x = -3$.

WORKED $\dfrac{8^{2/3} \cdot 2^{-1}}{4^{1/2}} = \dfrac{4 \cdot \frac{1}{2}}{2} = 1$

ACTIVITY 1

Spot the missing hypothesis

ACTIVITY 1 GROUPS OF THREE 12 MIN

Each line below is wrong as written, or is not defined for all real inputs. For each one, give a counterexample, then add a condition that makes a correct statement — or explain why no such simple repair works.

1. $(a^m)^n = a^{mn}$
2. $\sqrt{x^2} = x$
3. $\sqrt{ab} = \sqrt{a}\,\sqrt{b}$
4. $\log(a+b) = \log a + \log b$
5. $\dfrac{1}{a+b} = \dfrac{1}{a} + \dfrac{1}{b}$

Share: each group presents one repair and the counterexample that forced it.

The counterexample, then the condition

CLAIM	BREAKS AT	REPAIR, AND WHY
$(a^m)^n = a^{mn}$	$a = -1,\ m = 2,\ n = \frac{1}{2}$	Need $a > 0$.
$\sqrt{x^2} = x$	$x = -3$	$\sqrt{x^2} = \lvert x \rvert$ — the radical returns the non-negative root.
$\sqrt{ab} = \sqrt{a}\sqrt{b}$	$a = b = -1$	Need $a, b \geq 0$.
$\log(a+b) = \log a + \log b$	$a = b = 1$	**No repair.** The true law is $\log(ab) = \log a + \log b$.
$\dfrac{1}{a+b} = \dfrac{1}{a} + \dfrac{1}{b}$	$a = b = 1$	**No repair.** $\frac{1}{2} \neq 2$.

THE PATTERN Items 1–3 were true statements missing a hypothesis. Items 4–5 never were: no condition rescues them.

Absolute value is distance

DEFINITION

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Read $|a-b|$ as the distance from a to b on the line.

PROPOSITION For all real x, y and every $c > 0$:

$$|xy| = |x||y| \qquad |x| \leq c \iff -c \leq x \leq c$$

$$|x+y| \leq |x| + |y| \quad \text{(triangle inequality)}$$

WORKED $|2x-3| < 5$ says $2x$ is within 5 of 3, so $-2 < 2x < 8$, that is $-1 < x < 4$. Every step is an equivalence.

ACTIVITY 2

Distance on the line

ACTIVITY 2 PAIRS 10 MIN

Turn each sentence into an absolute-value inequality, solve it, and draw the solution set on a number line.

1. x is within 3 of 7.
2. x is at least 2 away from -1.
3. $2x$ is closer to 5 than 1 is.

Share: two pairs draw their number lines on the board; the room decides which endpoints are included, and says why.

The endpoints are the whole lesson

SENTENCE	INEQUALITY	SOLUTION	ENDPOINTS
x is within 3 of 7	$\lvert x-7\rvert \le 3$	$4 \le x \le 10$	Closed — "within" allows exactly 3 away.
x is at least 2 from -1	$\lvert x+1\rvert \ge 2$	$x \le -3$ or $x \ge 1$	Closed — "at least" includes equality.
$2x$ is closer to 5 than 1 is	$\lvert 2x-5\rvert < 1$	$2 < x < 3$	Open — "closer than" is strict.

WHY ITEM 2 SPLITS On an absolute value, "$\le$" gives one interval; "$\ge$" gives **two rays** — far from -1 means far on either side.

Three order rules explain the moves

RULES WE WILL USE For all real a, b, c:

1. exactly one of $a < b$, $a = b$, $b < a$ holds;
2. $a < b \implies a + c < b + c$;
3. $a < b$ and $0 < c \implies ac < bc$.

A USEFUL CONSEQUENCE Claim: $a < b \implies -b < -a$. By (2) add $-a - b$ to both sides of $a < b$:

$$a - a - b < b - a - b \implies -b < -a$$

No new inequality rule was needed: adding the same number, then reversing signs, was enough. ∎

Rules for working with inequalities

MOVE	RESULT	REASON
Add c to both sides	Same direction	Adding preserves order
Multiply by $c > 0$	Same direction	Positive multiplication preserves order
Multiply by $c < 0$	Reverse the sign	Negative multiplication reverses order
Multiply by an unknown x	Split into cases	The sign of x is unknown
Square both sides	Check the signs	$-3 < 2$ but $(-3)^2 > 2^2$

WORKED EXAMPLE Start with $-2 < 3$. Multiplying both sides by -4 gives 8 and -12. Because -4 is negative, reverse the sign: $8 > -12$.

CLASSIC From $\frac{1}{x} < 1$ you may **not** conclude $x > 1$. Check $x = -2$: the premise holds, the conclusion fails.

ACTIVITY 3

Always, sometimes, or never

ACTIVITY 3 GROUPS OF THREE 12 MIN

You are given $a < b$. Sort these moves into always legal, legal under a condition, or never legal. State the condition, or give a counterexample.

1. Add c to both sides.
2. Multiply both sides by c.
3. Square both sides.
4. Take reciprocals of both sides.
5. Multiply both sides by x, where x is the unknown.
6. Apply a strictly increasing function f to both sides.

Share: each group defends one card. Where two groups disagree, the counterexample settles it at the board.

Sorted, with the condition stated

MOVE ON $A < B$	VERDICT	CONDITION, OR THE COUNTEREXAMPLE
1 · Add c	Always	Rule (2), with no restriction on c.
2 · Multiply by c	Conditional	$c > 0$ keeps it; $c < 0$ reverses it; $c = 0$ destroys it.
3 · Square both sides	Conditional	Needs $0 \leq a$. Else $-3 < 2$ but $9 > 4$.
4 · Take reciprocals	Conditional	Needs a, b non-zero, *same* sign; then it reverses.
5 · Multiply by the unknown x	Never	Sign of x unknown — split into cases instead.
6 · Apply strictly increasing f	Always	That is the definition — and it explains cards 1 and 2.

Where the quadratic formula comes from

DERIVATION Start from $ax^2 + bx + c = 0$ with $a \neq 0$ and divide by a:

$$x^2 + \tfrac{b}{a}x = -\tfrac{c}{a}$$

Complete the square on the left, using $x^2 + \beta x = \left(x + \tfrac{\beta}{2}\right)^2 - \tfrac{\beta^2}{4}$:

$$\left(x + \tfrac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

If $b^2 - 4ac \geq 0$, take *both* square roots and subtract $\tfrac{b}{2a}$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

∎

WHAT THE DISCRIMINANT DECIDES $b^2 - 4ac > 0$: two real roots. $= 0$: one repeated root. < 0: none in $\mathbb{R}$ — which is exactly the gap Session 13 fills with $\mathbb{C}$.

ACTIVITY 4

Derive it without us

ACTIVITY 4 PAIRS 12 MIN

1. Solve $3x^2 - 12x + 7 = 0$ by completing the square. Do not use the formula.
2. In the general derivation, which step gives two possible answers? Explain why both must be kept.
3. Write one sentence saying what $b^2 - 4ac$ tells you, and why it tells you that.

Share: one pair derives the general case at the board while the room checks that every step is an equivalence.

Every step an equivalence

1 · COMPLETING THE SQUARE

Divide by 3:

$$x^2 - 4x + \tfrac{7}{3} = 0$$

Since $x^2 - 4x = (x-2)^2 - 4$:

$$(x-2)^2 = 4 - \tfrac{7}{3} = \tfrac{5}{3}$$

So $x = 2 \pm \dfrac{\sqrt{15}}{3}$.

2 · WHERE THE TWO ANSWERS APPEAR The square-root step:

$u^2 = k$ gives $u = \pm\sqrt{k}$, since both squares give k. Writing only $+\sqrt{k}$ discards a real solution.

3 · WHAT $B^2 - 4AC$ DECIDES It is the quantity under the root, so its sign decides whether a real root exists: > 0 two, $= 0$ one, < 0 none in $\mathbb{R}$.

EXIT TICKET · 10 MINUTES

On your own, before you leave

1 Solve $|3x+1| \geq 4$ and name which steps are equivalences.

2 Find all real x with $\frac{2}{x-1} \leq 1$, handling the sign of $x-1$ explicitly.

3 Show $x^2+y^2 \geq 2xy$ for all real x, y, and say exactly when equality holds.

RULE FOR TODAY One sentence of justification under every answer. Hand it in on the way out — it is not graded, it tells us who to sit with tomorrow.

WRAP-UP

What today was actually about

- → The rules you memorised are **theorems with hypotheses**. Learn the hypotheses.
- → Order has three axioms; every inequality move must trace back to them.
- → Marks live in the justification, not the answer.

hypothesis · counterexample · equivalence step · axiom · vacuous truth

Problem Set 1

12 diagnostic items — algebra, absolute value, inequalities. Each answer carries one sentence of justification.

Next session: **Sets & set operations** — the notation the whole course is written in.